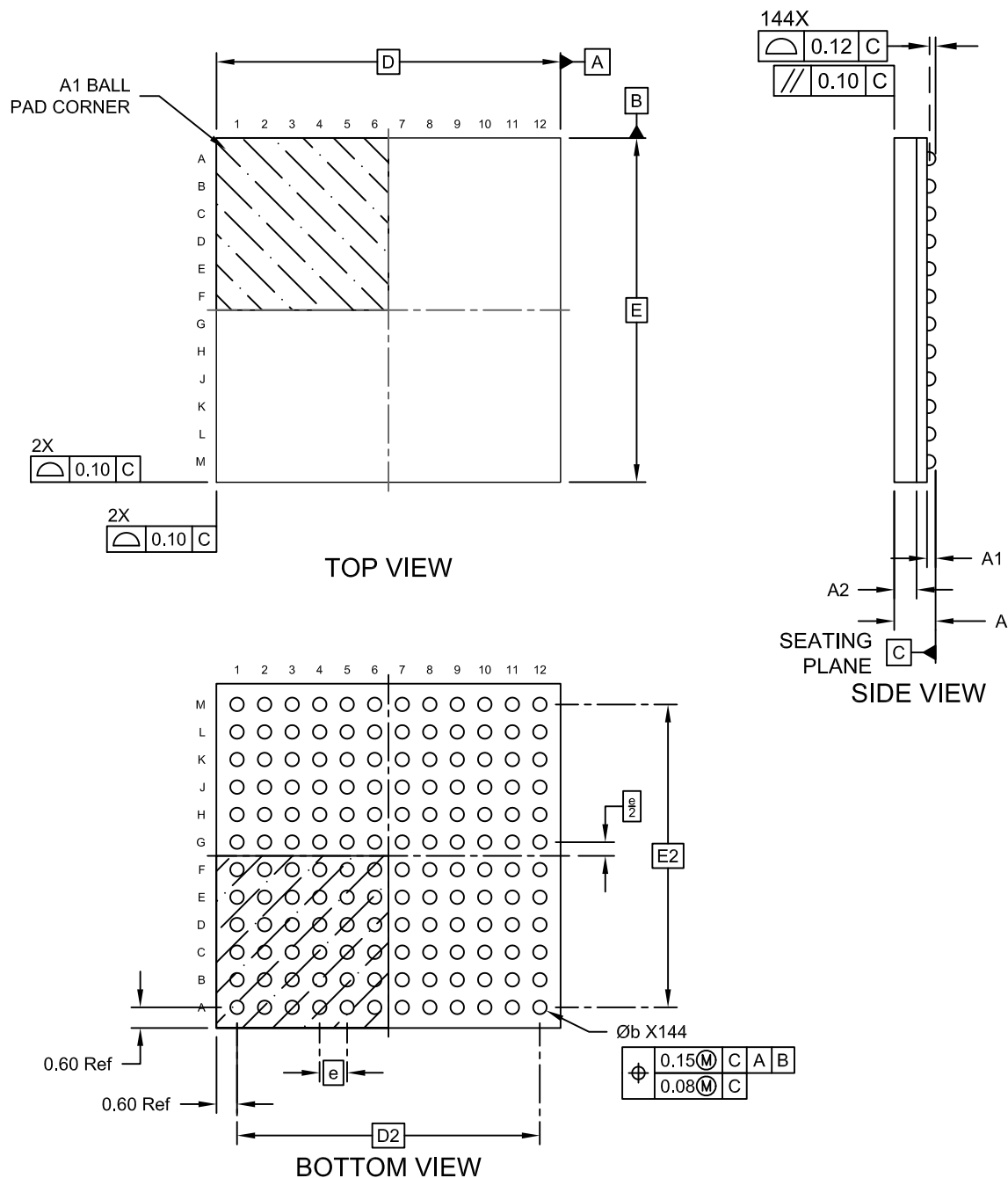


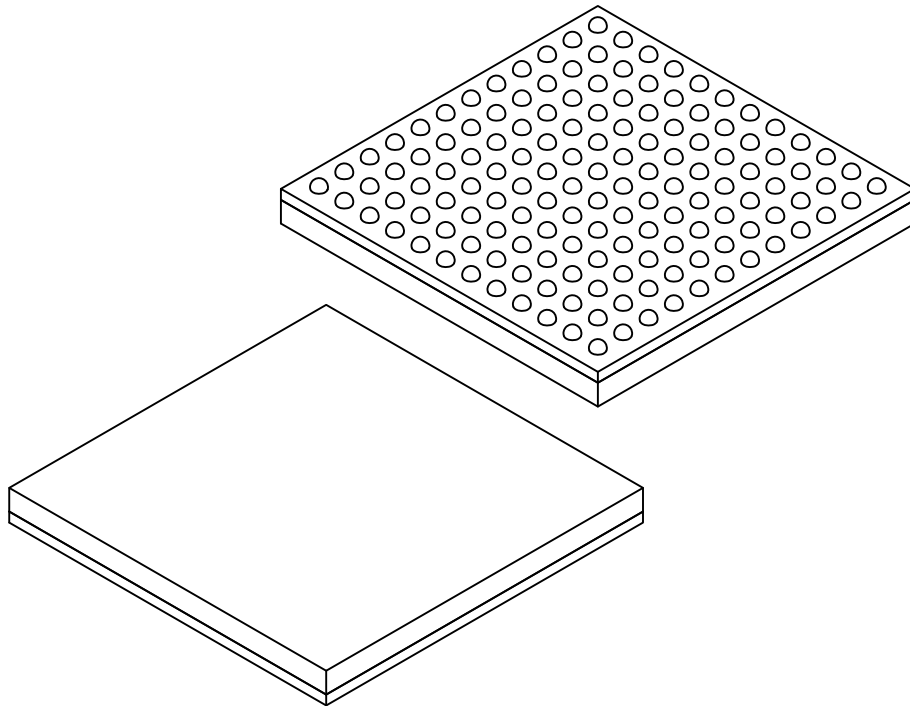
144-Lead Thin Fine-Pitch Ball Grid Array Package (B4B) - 10x10x1.2 mm Body [TFBGA] Atmel Legacy GPC: CGQ

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



144-Lead Thin Fine-Pitch Ball Grid Array Package (B4B) - 10x10x1.2 mm Body [TFBGA] Atmel Legacy GPC: CGQ

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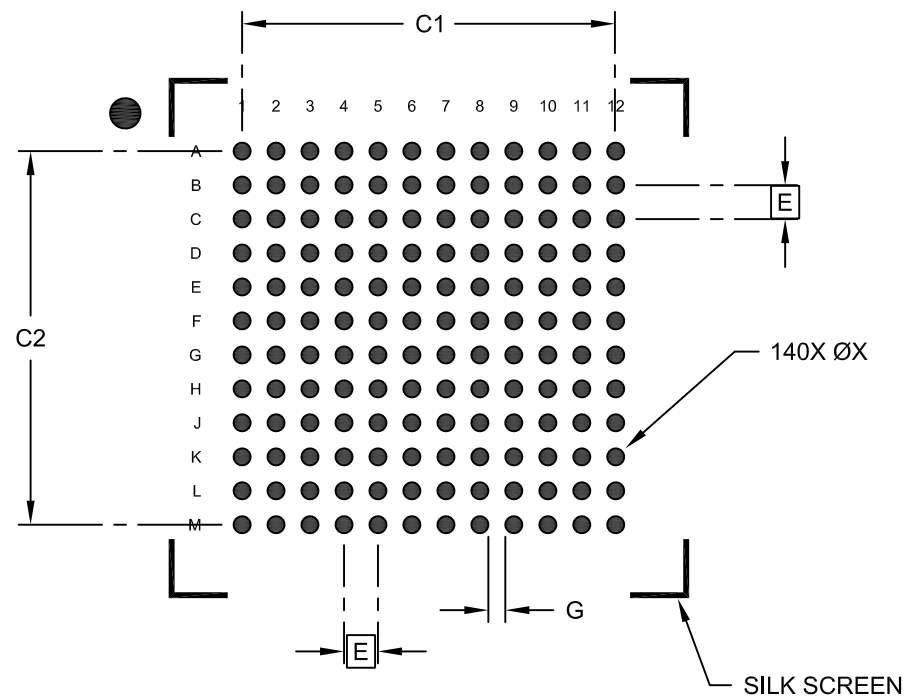
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	144		
Pitch	e	0.80 BSC		
Overall Height	A	-	-	1.20
Ball Height	A1	0.25	-	-
Mold Thickness	A2	0.55	-	-
Overall Length	D	10.0 BSC		
Ball Array Length	D2	8.80 BSC		
Overall Width	E	10.0 BSC		
Ball Array Width	E2	8.80 BSC		
Ball Diameter	b	-	0.40	-

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

144-Lead Thin Fine-Pitch Ball Grid Array Package (B4B) - 10x10x1.2 mm Body [TFBGA] Atmel Legacy GPC: CGQ

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Optional Pad Diameter (X247)	X			0.30
Contact Pad Spacing	C1		8.80	
Contact Pad Spacing	C2		8.80	
Column and Row Spacing	G	0.40		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23137 Rev A